STEP 5 — Dashboard, Demonstration aur Submission

Time: ~10 min · Is step ka output: live dashboard + demo script + submission pack

5.1 Dashboard chalana

```
streamlit run dashboard/app.py
```

Terminal me aisa dikhega:

```
  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://192.168.1.5:8501
```

Browser apne aap khul jayega. Na khule to `http://localhost:8501` manually kholiye.

Band karne ke liye: terminal me `Ctrl + C` daba dijiye.

> **Important:** dashboard chalane se pehle `python run_all.py` chal chuka hona chahiye. Warna screen par red error aayega: “Processed data not found. Run the pipeline first.”

5.2 Dashboard me kya-kya hai

Sidebar filters (left taraf)

Filter	Kya karta hai
Care pathway	Sirf chuni hui pathways ka data dikhata hai
Department	Sirf chune hue department
Length of stay slider	LOS ki range set karta hai
Triage level	Emergency ki severity ke hisaab se filter

Har filter turant sab charts, KPI cards aur tables update kar deta hai — yeh dashboard ki sabse acchi baat hai, demo me isko highlight kijiye.

5 KPI cards (top par)

KPI	Matlab
Patient cases	Kitne patients (filter lagne ke baad)
Total events	Kitni activities record hui
Avg length of stay	Average kitne ghante hospital me rahe

Avg cost / case	Per patient average kharcha
Avg conformance fitness	Real cases model se kitna match karte hain

5 Tabs

Tab	Andar kya hai
Overview	LOS histogram + long-stay threshold line, monthly volume line chart, bottleneck bar chart, department cost treemap, resource workload, pathway scatter, auto-generated insights list
Process discovery	DFG diagram, top-10 variants, 4 metric cards (cases / activities / edges / variants), poori summary JSON
Conformance	Fitness metric cards, fitness distribution, deviation bar chart, variant-wise conformance table
ML model	Best model naam, saare models ka comparison table, ROC curve, confusion matrix, feature importance chart + table
Data	Processed event log (300 rows preview) + filtered CSV download button

5.3 Screenshot lena (report me daalne ke liye)

1. Dashboard kholiye, sab filters default rakhiye
2. **Windows:** `Win + Shift + S` → area select kijiye. **Mac:** `Cmd + Shift + 4`
3. Yeh 4 screenshots zaroor lein:

#	Kya capture karna hai	Report me kahan
1	Poora Overview tab (KPI cards + 3-4 charts)	Results → Dashboard section
2	Process discovery tab (DFG diagram)	Results → Process model
3	Conformance tab (fitness metrics)	Results → Conformance
4	ML model tab (ROC + feature importance)	Results → Prediction

Screenshots `docs/screenshots/` folder me save kar lijiye.

5.4 2-minute demo script (bol-bol ke practice kijiye)

[0:00 – 0:20] Problem
“Hospital me ek patient ko Registration se Discharge tak kai steps se guzarna padta hai. Real me har patient ka rasta alag hota hai. Sirf report dekhne se pata nahi chalta ki kahan wait ho raha hai aur kahan kharcha badh raha hai.”

[0:20 – 0:40] Solution
“Maine CareFlow banaya — ek clinical pathway process mining system. Yeh hospital ke event log ko padhta hai — har row matlab ek patient ka ek step — aur usse asli process model, bottlenecks, conformance aur prediction nikalta hai.”

[0:40 – 1:10] Discovery demo
“Yeh hai hamara discovered process model — Directly-Follows Graph, 28 edges, 15 activities. 1,500 patients ke 13,583 events se bana hai. 14 different pathway variants mile, lekin top 5 variants milkar 81 % cases cover karte hain — matlab zyada tar patients standard pathway follow karte hain.”

[1:10 – 1:40] Insights demo
“Yeh Overview tab. Average stay 9.65 ghante. Sabse bada bottleneck hai **Observation** — average 226 minute wait. Costliest department Operation Theatre hai. Poore 6 mahine ka treatment cost 2.58 crore rupaye. Load Friday 2 baje peak par hai — matlab staff roster usi hisaab se lagana chahiye.”

[1:40 – 2:00] Prediction demo
“Aur yeh predictive part — main ek model train kiya jo admission ke waqt hi bata deta hai ki patient ka stay 24 ghante se zyada ho sakta hai ya nahi. 4 real-world drivers mile: kitne events hue, rework kitna hua, triage level, aur admission type. Isse hospital pehle se plan kar sakta hai.”

5.5 Viva / review me poochhe jaane wale 12 sawaal

#	Sawaal	Aapka jawab (short)
1	Process mining kya hai?	Event logs se asli business process ko discover, monitor aur improve karne ki technique. 3 hisse: discovery, conformance, enhancement.
2	Event log me minimum kya chahiye?	Case ID, activity, timestamp — “process mining ke teen zaroori columns”.
3	DFG aur Petri net me farak?	DFG sirf “kaun kis ke baad aaya” dikhata hai. Petri net me places aur tokens hote hain, isliye conformance properly check ho jaata hai.
4	Variant kya hota hai?	Ek case ka poora activity sequence. Same sequence wale cases ek hi variant ke hain.
5	Conformance checking kaise kiya?	Token-based replay — actual trace ko discovered model par chalaya, missing aur remaining tokens gine, usse fitness nikala.
6	Fitness 100 % kyun aaya?	Kyunki data synthetic hai aur generator ke saare paths model me cover hote hain. Real data par 0.70–0.95 aata hai. **(honest answer)**
7	Data leakage kya hai, aapne kaise handle kiya?	Outcome ke baad bante columns (waiting time, service time) model se hata diye — production model leakage-free features par bana hai, tabhi results honest hote hain.
8	Imbalanced data kaise handle kiya?	Long-stay cases sirf 1.13 % hain, isliye accuracy ki jagah **F1, Precision, Recall, ROC-AUC** par model select kiya, aur stratified 80/20 split + 5-fold cross-validation use kiya.
9	Kaunsa model best nikla?	Leakage-free set par Gradient Boosting — accuracy 99.7 %, F1 0.80, ROC-AUC 0.998.
10	Business impact kya hai?	Bottleneck (Observation wait) ka pata chala, staff roster peak hour ke hisaab se plan ho sakta hai, aur long-stay patients ko pehle se flag karke bed planning ho sakti hai.

11	Project ki limitation?	Reference data synthetic hai; asli hospital data ke liye ETL aur data-privacy clearance chahiye. Model retraining bhi regular chahiye.
12	Aage kya kar sakte hain?	Real HIS/EMR integration, live alerting, simulation (what-if), aur object-centric process mining multiple patients ke saath.

5.6 Submission pack — kya-kya bhejna hai

Assignment requirement	File	Notes
Source Code (if applicable)	`CareFlow_Project.zip`	Poora project + data + outputs + docs
Project Documentation	`CareFlow_Project_Documentation.pdf`	Technical documentation
Project Report	`CareFlow_Project_Report.pdf`	Complete report with charts
Presentation (PPT)	`CareFlow_Presentation_Content.pdf`	13 slides — PowerPoint me copy-paste kijiye
Model Evaluation Results	`CareFlow_Model_Evaluation_Results.pdf`	Saare metrics + leakage analysis
Final Project Demonstration	`CareFlow_Step5_Dashboard_And_Demo.pdf`	Demo script + live dashboard

Email draft (mentor ko bhejne ke liye)

```
Subject: CareFlow - Clinical Pathway Process Mining - Final Project Submission

Respected Ms. Jyotirmayee Behera,

I am submitting the final deliverables for my internship project
"CareFlow: Clinical Pathway Process Mining" (Ref. No. ITS/DA/1393,
Data Analytics domain, Data Associate L1 Intern).

Attached:
1. Source Code            - CareFlow_Project.zip
2. Project Documentation  - CareFlow_Project_Documentation.pdf
3. Project Report         - CareFlow_Project_Report.pdf
4. Presentation (PPT)     - CareFlow_Presentation_Content.pdf
5. Model Evaluation       - CareFlow_Model_Evaluation_Results.pdf
6. Final Demonstration    - careFlow Dashboard (streamlit run dashboard/app.py)

Project summary:
- 1,500 patient cases, 13,583 events, 15 clinical activities
- Discovered process model: 28 directly-follows edges, 14 pathway variants
- Average length of stay 9.65 h; biggest bottleneck Observation (226.6 min wait)
- Conformance analysis + long-stay prediction model (Gradient Boosting,
  ROC-AUC 0.998 on leakage-free features)
- Interactive Streamlit dashboard with 5 analysis tabs

All work is original. Please let me know if any clarification is required
or if a demonstration session is needed.
```

```
Regards,
Yash Rahangdale
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```

5.7 Final checklist (submit karne se pehle)

✔	Kaam
☐	`python run_all.py` clean chala aur “Pipeline finished” dikha
☐	`outputs/figures/` me 14 PNG files hain
☐	`outputs/models/best_model.joblib` maujood hai
☐	`streamlit run dashboard/app.py` khulta hai aur saare 5 tabs kaam karte hain
☐	Dashboard ke 4 screenshots le liye
☐	Report me report me aapka naam, ref no. aur dates sahi hain
☐	PPT me screenshots aur real numbers daal diye
☐	Demo script bolke practice kar li (2 minute me)
☐	Viva ke 12 sawaal padh liye
☐	ZIP bana liya aur sab PDFs attach kar diye

Mubarak ho — project complete hai! Poora CareFlow: data se dashboard tak, source code se report tak, sab ready.

Ab aapka kaam: **run kijiye, samajh lijiye, aur confidently present kar dijiye.**